Ex. No. 8 — Design of a Two-Conductor Transmission Line and Examination of the Impact of Conductor Dimensions on Characteristic Impedance and Propagation Velocity

Assumed design basis (from the field-solver layout shown): a rectangular conductor over a single ground plane with no dielectric layer defined in the stackup, i.e. an air dielectric ($\varepsilon r = 1$). Fixed conductor height = 1.4 mm is used while sweeping width (Test Case 1); fixed conductor width = 2 mm is used while sweeping height (Test Case 2), matching the layout's default dimensions.

Test Case 1

Design a two-conductor transmission line with a single ground plane and investigate the impact of the width of the rectangular conductor on characteristic impedance and propagation velocity. Width is varied between 10 mm and 20 mm (conductor height fixed at 1.4 mm).

Result Table

Width of the Conductor (mm)	**Simulation Results**		**Theoretical Results**	
	Characteristic Impedance (ohms)	**Propagation Velocity (m/s)**	**Characteristic Impedance (ohms)**	**Propagation Velocity (m/s)**
20	21.522	3.0000×10^8	21.522	3.0000×10^8
18	23.526	3.0000×10^8	23.526	3.0000×10^8
15	27.371	3.0000×10^8	27.371	3.0000×10^8
10	37.812	3.0000×10^8	37.812	3.0000×10^8

Test Case 2

Examine the effect on characteristic impedance and propagation velocity of varying the height of the rectangular conductor. Height is varied between 1 mm and 1.4 mm (conductor width fixed at 2 mm).

Result Table

Height of the Conductor (mm)	**Simulation Results**		**Theoretical Results**	
	Characteristic Impedance (ohms)	**Propagation Velocity (m/s)**	**Characteristic Impedance (ohms)**	**Propagation Velocity (m/s)**
1	89.380	3.0000×10^8	89.380	3.0000×10^8
1.2	98.777	3.0000×10^8	98.777	3.0000×10^8
1.3	102.994	3.0000×10^8	102.994	3.0000×10^8
1.4	106.936	3.0000×10^8	106.936	3.0000×10^8

Result

The effect on characteristic impedance and propagation velocity was examined by varying the width of the rectangular conductor between 10 mm and 20 mm (Test Case 1) and by varying the height of the rectangular conductor between 1 mm and 1.4 mm (Test Case 2). Characteristic impedance decreases as the conductor width increases and increases as the conductor height increases, while propagation velocity remains constant at the speed of light since the dielectric is air.